# Q: Define candidate key and state its two fundamental properties in the context of relational databases.

A candidate key is a minimal super key for a relation.

A set of attributes forms a candidate key if it satisfies two fundamental properties;

1. **Uniqueness;**

   No two distinct tuples in any valid relation state can have the same combination of values for these attributes.

| A | B | C |
|---|---|---|
| 1 | 2 | 5 |
| 2 | 3 | 6 |
| 3 | 4 | 7 |

→ (A,B) can be a candidate key (unique for each tuple)

2. **Irreducibility (Minimality):**

   No proper subset of these attributes can possess the uniqueness property. If any attribute is removed from the candidate key, the remaining attributes no longer uniquely identify the tuples.

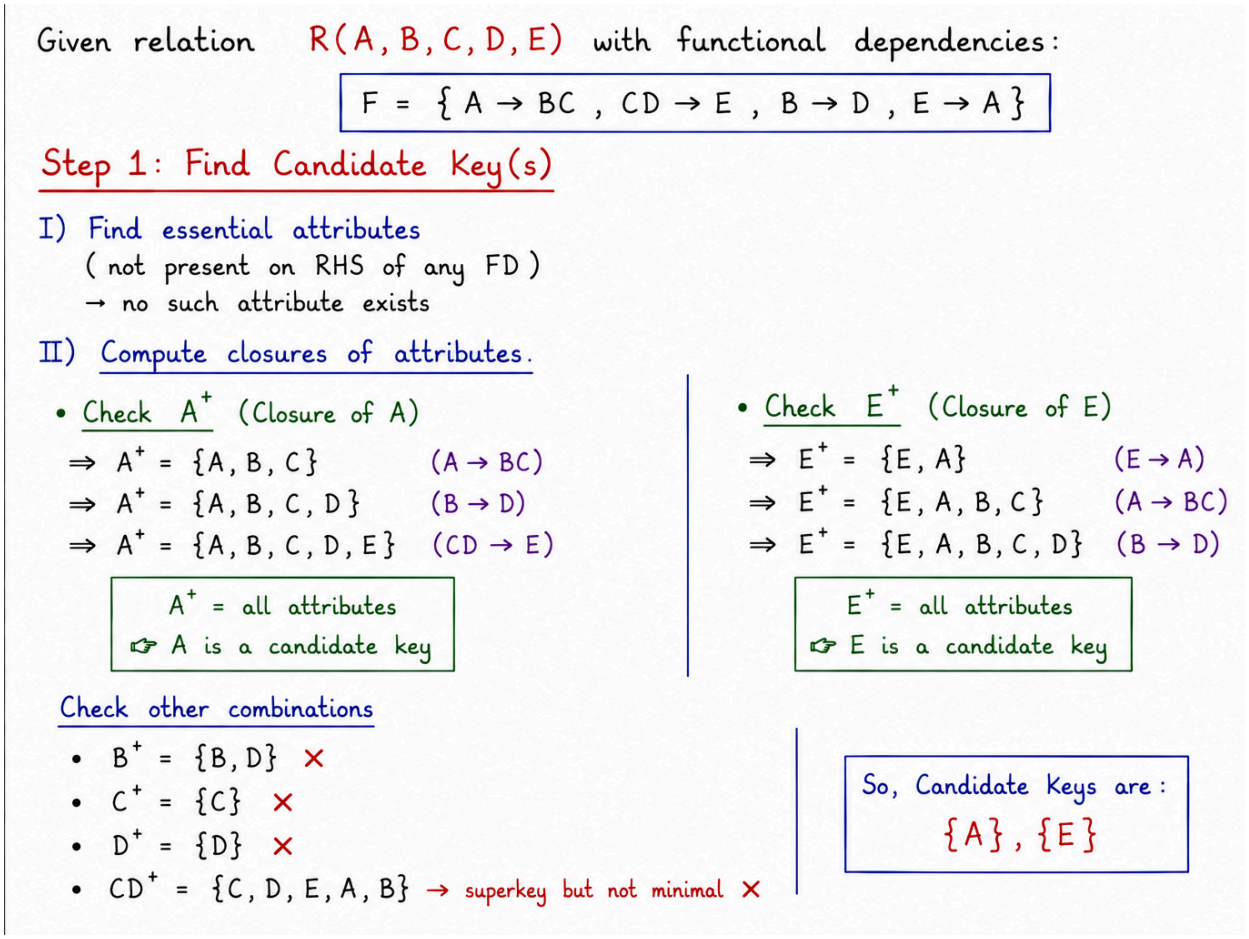
Example: (A,B) is a candidate key

→ (A) alone → Not unique

→ (B) alone → Not unique

} Hence, (A,B) is minimal

**Q - Consider a relation R(A, B, C, D, E) with the set of functional dependencies F = {A -> BC, CD -> E, B -> D, E -> A}. Find the candidate key(s) for relation R and normalize the relation up to Third Normal Form (3NF), showing each step clearly.**




**Step 2: Check for 1NF**

• The relation contains atomic attributes, so it is in 1NF.

## Step 3: Check for 2NF

- A relation is in 2NF if it is in 1NF and every non-prime attribute is fully functionally dependent on any candidate key.
- Prime attributes: A, E.
  Non-prime attributes: B, C, D
- Check dependencies:
  - ✳ A → BC (A is a candidate key, full dependency).
  - ✳ CD → E (no partial dependency as CD is not a subset of candidate key)
  - ✳ B → D (no partial dependency as B is not a subset of candidate key)
  - ✳ E → A (E is a candidate key, full dependency).
- Therefore, the relation is in 2NF.

# Step 4: Check for 3NF

- A relation is in 3NF if it is in 2NF and for every functional dependency $X \to Y$, at least one of the following holds:
  1. $X \to Y$ is a trivial FD
  2. $X$ is a super key.
  3. $Y$ is a prime attribute (part of a candidate key).

| FD | 3NF? | Reason |
|---|---|---|
| $A \to BC$ | ✓ | A is a candidate key |
| $CD \to E$ | ✓ | E is prime |
| $B \to D$ | ✗ | B is not a key and D is not prime |
| $E \to A$ | ✓ | E is a candidate key |

∴ Since $B \to D$ violates 3NF, the relation is not in 3NF.

- **Decomposition to achieve 3NF:**

For each violating FD:

→ Create a new relation with (LHS ∪ RHS)

For the violating FD: $B \to D$

Create a new relation: $R_1(B, D)$

This relation is in 3NF because:
$B \to D$ and $B$ is the key of $R_1$.

– – Remove RHS attributes of violating FD (D) from the original relation

Remaining Relation: $R_2(A, B, C, E)$

Now check dependencies in $R_2$:
- $A \to BC$
- $E \to A$

} A and E are candidate keys, so $R_2$ is in 3NF.

Original Relation

$R(A, B, C, D, E)$

↓

Decomposed Relations

$R_1(B, D)$

Key: B
FD: $B \to D$
(3NF)

$R_2(A, B, C, E)$

FDs:
$A \to BC$
$E \to A$
(3NF)

## Dependency Preservation ???

A decomposition is dependency-preserving if all the functional dependencies of the original relation can be checked without joining the decomposed relations.

Original FDs for given example – $F = \{A \rightarrow BC,\ CD \rightarrow E,\ B \rightarrow D,\ E \rightarrow A\}$

Decomposed Relations: $R_1(B, D)$ $R_2(A, B, C, E)$

Map FDs to Relations

| FD | Mapped To | Preserved? |
|---|---|---|
| $B \rightarrow D$ | $R_1(B, D)$ | ✓ preserved |
| $A \rightarrow BC$ | $R_2(A, B, C, E)$ | ✓ preserved |
| $E \rightarrow A$ | $R_2(A, B, C, E)$ | ✓ preserved |
| $CD \rightarrow E$ | C, D and E are not all in one relation | ✗ not preserved |

Therefore: ✗ Not Dependency Preserving

# Rule for Lossless Join

For decomposition of R into $R_1$ and $R_2$:

A decomposition is lossless if the common attributes between $R_1$ and $R_2$ determine all attributes of at least one relation.

## Formal Rule

If:

$$R_1 \cap R_2 \rightarrow R_1$$

OR

$$R_1 \cap R_2 \rightarrow R_2$$

Then:

✓ Decomposition is Lossless Join

## Lossless Join ??

A decomposition is **lossless** if joining the decomposed relations produces exactly the original relation without losing information or creating extra tuples.

Decomposition:

$R_1(B, D)$ $R_2(A, B, C, E)$

Common attribute:

$R_1 \cap R_2 = \{B\}$

We have:

$B \rightarrow D$

Therefore:

$B \rightarrow R_1(B, D)$ ⟶ So the common attribute B determines all attributes of $R_1$.

✓ Therefore, the decomposition is Lossless Join.

### How to Test for Lossless Join (Step-by-Step)

1. **Find common attributes:** Identify $R_1 \cap R_2$.
2. **Find the attribute closure:** Compute $(R_1 \cap R_2)^+$ using the set of Functional Dependencies (FDs).
3. **Verify superkey status:**
   - If $(R_1 \cap R_2)^+$ contains all attributes of $R_1$, the decomposition is **lossless**.
   - If $(R_1 \cap R_2)^+$ contains all attributes of $R_2$, the decomposition is **lossless**.
   - If neither condition is met, the decomposition is **lossy**.

### Quick Example

Given $R(A, B, C)$ with functional dependency $A \rightarrow B$, decompose $R$ into:

- $R_1(A, B)$
- $R_2(A, C)$

1. **Intersection:** $R_1 \cap R_2 = \{A\}$
2. **Closure of shared attribute:** $A^+ = \{A, B\}$
3. **Check:** $\{A, B\}$ contains all attributes of $R_1(A, B)$. Therefore, $A$ is a key for $R_1$, making this a **lossless join**.

## MODULE - 1

Q1. Explain DBMS and file processing systems. Compare DBMS with file systems with respect to advantages, limitations, examples, uses and applications.

Q2. Explain the three-level architecture of DBMS with a neat diagram. Explain the external, conceptual and internal levels and discuss logical and physical data independence with examples.

Q3. Explain the main characteristics of a database system. How do these characteristics help in reducing redundancy, maintaining consistency, sharing data and providing security?

Q4. Explain the different types of database users and their roles, including DBA, database designers, system analysts, application programmers, naive users and sophisticated users.

Q6. Explain the role of databases in healthcare. Give suitable examples of their uses and applications in hospitals or healthcare management systems.

Q7. Explain the basic concepts of ER modelling: entity, entity set, attributes and relationship. Explain different types of attributes with suitable examples.

Q8. Explain relationship and relationship set in ER modelling. Describe cardinality and participation constraints (1:1, 1:N, M:N, total and partial participation) with suitable examples.

Q9. Explain super key, candidate key, primary key, alternate key and foreign key with examples. Also explain domain, entity integrity and referential integrity constraints.

Q10. Explain specialization and generalization in ER modelling with suitable examples and diagrams. Differentiate between them and explain their constraints.

Q12. Design an ER diagram for a Hospital Management System. Identify entities, attributes, keys, relationships, cardinality, participation constraints and any suitable specialization/generalization.

## MODULE - 2

Q1. Explain the Relational Model and its basic terminology, including relation, tuple, attribute, domain, degree, cardinality, relation schema and relational database. Give suitable examples.

Q2. Explain how an ER/EER diagram is converted into a relational schema. Explain the mapping of entities, attributes, relationships, weak entities and specialization/generalization with suitable examples.

Q3. What is Relational Algebra? Explain its basic operations with suitable examples: Selection, Projection, Union, Set Difference, Cartesian Product and Rename.

Q5. What is normalization? Explain the need for normalization with suitable examples. What are functional dependencies? Explain functional dependency with examples and identify functional dependencies from a given relation.

Q6. Explain normalization and the normal forms 1NF, 2NF, 3NF, BCNF and 4NF with suitable examples.

Q7. Explain decomposition in normalization. Discuss the rules/properties of decomposition, including lossless join and dependency preservation, with suitable examples.